MIT Teacher Uses AI to Help Data Centers Use Less Power

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Data centers are big buildings full of computers. They use lots of **electricity** and can make power grids work harder. Christina Delimitrou, a professor at MIT, wants to fix that by making the computers inside run smarter.

Her team uses **machine learning** so cloud systems waste less energy. They improve old **software**, share hardware more carefully, and design leaner servers. That way, companies can do more computing with the machines they already have.

Delimitrou says if **data centers** are not used well, they burn more power than they need. Cutting extra **software** "bloat" without slowing things down could mean fewer new **data centers**.

Earlier research found many large systems ran at only about 15 percent **capacity**. **Machine learning** can help manage thousands of machines better. Her work shows smarter **software**not only more buildingscan meet growing internet demand.

Word Treasure

data centers -- Big buildings full of computers that power the internet.
machine learning -- A kind of AI that learns patterns from data.
electricity -- Power that runs computers, lights, and machines.
software -- Programs that tell computers what to do.
capacity -- How much work machines can do when fully used.

Background you need

Data centers are buildings packed with computers that run websites and apps.

Those computers need lots of electricity to work and stay cool.

Smarter software can help the same machines do more work with less wasted power.

Break it down

WHO: MIT professor Christina Delimitrou and her team
WHAT: Using AI to make data centers more efficient
WHERE: Research at MIT on large cloud computer buildings
WHY: Data centers use lots of power and can waste energy

Why it matters

Apps kids use run in data centers, so saving electricity there helps the planet.

Test what you remember

1. Where does Christina Delimitrou work?

(A) The Moon base
(B) A farm only
(C) A candy shop
(D) MIT

2. What do data centers use a lot of?

(A) Paper airplanes
(B) Electricity
(C) Sandcastles
(D) Snowballs

3. What tool helps make the systems smarter?

(A) Machine learning
(B) A whistle
(C) A broom only
(D) A kite

4. About what capacity did many systems run at?

(A) About 15 percent
(B) Zero forever
(C) Exactly one percent always
(D) 200 percent

5. What could removing software bloat mean?

(A) Books disappear
(B) Rain stops
(C) Fewer new data centers may be needed
(D) All phones stop working

6. What else can AI help programmers do here?

(A) Fly without planes
(B) Grow trees overnight
(C) Bake bread with no oven
(D) Find and fix problems in cloud apps

Answer key (try the quiz first!): 1: D 2: B 3: A 4: A 5: C 6: D

Questions to think about

1. Christina Delimitrou: Says underused data centers burn extra power and smarter software can help.

MY THOUGHTS (at least 20 words):